

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

88yGyDdvVRHcVsZF6obPaV3Qov7oX4L1HzAA15uboop

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Compounds: Amoxicillin, Aspirin, Bismuth Subsalicylate, Ecstasy

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: Amoxi-Bism-Salylate

Formula: C28H31N3O9S

Weight: 605.63

Drug-likeness: 0.32

Activity:

Antibacterial and anti-inflammatory

Target Analysis

Bacterial Penicillin-Binding Proteins

Binding Affinity: -8.2

Essential for the antibacterial activity, disrupting bacterial cell wall synthesis.

Cyclooxygenase Enzymes (COX-1, COX-2)

Binding Affinity: -7.5

Associated with anti-inflammatory actions through the inhibition of prostaglandin synthesis.

Toxicity Profile

Risk Level: Moderate

Affected Systems: Liver, GI Tract, Kidneys

Mitigation Suggestions:

- Use hepatoprotective agents
- Monitor renal function
- Hydration

Pharmacokinetics

Absorption:

Moderate oral bioavailability, dependent on food intake

Distribution:

Wide tissue distribution with plasma protein binding of 60%

Metabolism:

Primarily hepatic through carboxylation and nitration

Excretion:

Renally excreted with unchanged metabolites

Half-life: 4-6 hours

Detailed Analysis

The newly hypothesized compound Amoxi-Bism-Salylate leverages the potent antibacterial properties of Amoxicillin combined with the anti-inflammatory effects of Aspirin. Bridging the effects with Bismuth's gastrointestinal protective and antimicrobial actions, the novel compound offers a multifaceted therapeutic approach. Targeting bacterial Penicillin-Binding Proteins ensures effective antibacterial action, while COX enzyme inhibition addresses inflammation. Theoretically, this compound could be pivotal in addressing infections accompanied by significant inflammatory responses. However, the potential toxicity, especially affecting the liver and kidney, necessitates careful consideration, dose optimization, and additional metabolic studies to improve its safety profile. Supporting the hypothesis, the development strategy emphasizes balancing efficacy against adverse effects, with a clinical trial designed to elucidate both therapeutic potential and safety comprehensively.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

<https://ALCHEMUS.ai>